

KHARGA, Egypt

WMO: 624350

Lat: 25.4644N

Lon: 30.5425E

Elev: 73

StdP: 100.45

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.7	(c) 7.0	(d) -3.3	(e) 2.9	(f) 15.6	(g) -1.7	(h) 3.3	(i) 15.5	(j) 9.5	(k) 17.9	(l) 8.8	(m) 18.9	(n) 0.8	(o) 280	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.9	(c) 44.0	(d) 22.8	(e) 42.6	(f) 22.5	(g) 41.4	(h) 22.2	(i) 24.9	(j) 39.9	(k) 24.1	(l) 39.1	(m) 23.4	(n) 38.4	(o) 3.4	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.8	34.0	18.9	13.8	33.5	17.9	12.9	33.6	76.1	40.0	72.7	39.0	69.9	38.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.4	(b) 8.4	(c) 7.5	(e) 2.8	(f) 46.7	(g) 1.3	(h) 1.6	(i) 1.9	(j) 47.8	(k) 1.1	(l) 48.7	(m) 0.4	(n) 49.6	(o) -0.6	(p) 50.8		
			DB													
			WB													
				2.8	46.7	1.3	1.6	1.9	47.8	1.1	48.7	0.4	49.6	-0.6	50.8	
				1.2	26.0	1.6	1.2	0.1	26.9	-0.9	27.6	-1.8	28.3	-3.0	29.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	15.2	17.3	21.1	26.1	30.4	32.8	33.2	33.5	31.5	27.5	21.6	16.7
	DBStd	7.25	2.68	3.29	3.67	3.74	3.37	2.50	1.99	1.94	2.44	3.16	3.27	2.66
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	243	105	52	12	1	0	0	0	0	0	0	7	67
	CDD10.0	5703	161	206	346	483	632	684	721	728	646	543	347	207
	CDD18.3	2904	8	25	99	233	374	434	462	469	396	285	104	15
	CDH23.3	43105	113	329	1151	2996	5584	7082	7748	7813	5873	3345	922	147
CDH26.7	27413	22	117	522	1683	3569	4829	5383	5426	3713	1777	334	38	
Wind	WSAvg	3.0	2.3	2.7	3.1	3.4	3.4	3.7	2.9	3.0	3.6	3.4	2.8	2.4
Precipitation	PrecAvg	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecMax	6	3	1	0	2	2	0	0	0	3	0	2	0
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	1	1	0	0	0	0	0	0	0	1	0	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	33.8	38.4	42.7	45.5	45.6	45.1	44.5	43.4	40.6	35.4	30.5
		MCWB	17.3	18.6	19.6	21.8	22.2	22.8	23.4	23.1	23.4	22.3	19.7	18.0
	2%	DB	26.5	30.1	35.0	39.9	42.8	43.5	43.1	42.7	41.2	38.0	32.6	27.1
		MCWB	15.5	16.7	18.0	20.7	21.6	22.4	23.0	22.9	22.6	21.5	19.1	16.6
	5%	DB	23.9	27.3	32.1	37.5	40.8	41.9	41.7	41.4	39.6	36.1	30.2	24.6
		MCWB	14.1	15.4	16.9	19.5	21.2	21.9	22.6	22.7	22.2	21.0	17.9	15.0
	10%	DB	21.8	24.7	29.2	34.9	38.7	40.2	40.5	40.3	38.0	34.2	28.2	22.7
		MCWB	13.3	14.3	16.0	18.3	20.5	21.4	22.2	22.4	21.8	20.1	17.1	14.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	19.5	20.5	23.3	24.4	25.4	25.9	26.0	25.2	23.8	21.3	19.5
		MCDB	26.9	31.7	35.8	39.2	41.1	41.8	41.9	39.7	39.4	37.2	32.3	28.7
	2%	WB	16.1	17.3	18.8	21.4	22.9	23.6	24.5	24.9	24.0	22.6	19.7	17.0
		MCDB	24.8	28.6	32.7	37.5	39.6	39.8	40.2	38.9	38.1	35.2	30.7	25.9
	5%	WB	14.8	15.8	17.5	20.2	21.9	22.7	23.7	24.0	23.1	21.6	18.6	15.6
		MCDB	23.1	26.0	30.8	35.7	38.6	39.2	39.2	38.5	37.3	33.9	28.9	23.4
	10%	WB	13.5	14.6	16.4	18.9	20.9	22.0	22.9	23.2	22.3	20.7	17.6	14.5
		MCDB	21.1	24.1	28.4	33.5	37.5	38.5	38.5	37.9	36.2	32.6	27.3	22.1
Mean Daily Temperature Range	5% DB	MDBR	13.5	13.9	14.6	15.1	14.7	14.1	14.4	13.9	13.1	12.8	13.0	13.0
		MCDBR	16.5	17.6	19.1	18.9	17.7	16.3	16.5	15.2	14.7	14.5	15.0	15.5
		MCWBR	8.9	8.5	8.2	7.8	7.0	5.8	6.2	5.6	5.4	5.6	6.8	8.0
	5% WB	MCDBR	15.8	16.6	17.6	17.9	16.5	15.3	15.3	13.8	13.8	13.7	14.1	14.7
		MCWBR	9.0	8.7	8.1	8.0	7.0	6.0	6.3	5.6	5.7	5.9	6.9	8.0
Clear-Sky Solar Irradiance	taub		0.386	0.423	0.473	0.567	0.585	0.513	0.505	0.501	0.491	0.485	0.410	0.389
	taud		2.057	1.936	1.811	1.628	1.611	1.780	1.819	1.837	1.844	1.841	2.007	2.050
	Ebn at Noon		849	846	824	757	742	792	798	801	797	773	816	827
	Edh at Noon		146	178	213	262	266	223	214	210	203	193	153	142
All-Sky Solar Radiation	RadAvg		4.55	5.46	6.54	7.36	7.80	8.13	7.90	7.49	6.76	5.71	4.80	4.27
	RadStd		0.12	0.19	0.21	0.13	0.17	0.08	0.09	0.08	0.15	0.10	0.09	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.30	N/A	N/A	N/A	-0.95	-1.70	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.30	N/A	+1.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page